

An inequivalent GNS fiber has no residual common range

MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_zero

theorem

Uses compression and cyclic transport to exclude fixed vectors in every other chosen pure-state class.

Statement

Fix a shell family and indices $i, j \in I$ as below, with $j \neq i$. For $f_{i,n} = \alpha_i(f_n)$, let $F_j = \bigcap_{n \geq 0} \text{ran } \pi_j(f_{i,n})$ and let P_j be its orthogonal projection. Then $P_j = 0$, or equivalently $F_j = \{0\}$.

Assumptions

Let A be the completed CAR algebra, the completion of the matrix stages $M_{2^n}(\mathbb{C})$ under $a \mapsto a \otimes I_2$. Write f_n for the image of the first diagonal matrix unit. These are decreasing projections with $f_0 = 1$. Write ι_n for the canonical embedding of stage n . The root state ω is determined by $\omega(\iota_n(a)) = a_{00}$ on every stage.

Choose one pure state φ_j from each unitary-equivalence class $j \in I$ of pure-state Gelfand-Naimark-Segal (GNS) representations, retaining ω at the root index o . Write (H_j, π_j, ξ_j) for its complete complex GNS space, unital star representation and unit cyclic vector. Thus $\varphi_j(a) = \langle \xi_j, \pi_j(a) \xi_j \rangle$, and the vectors $\pi_j(a) \xi_j$ are dense in H_j . Inner products are linear in the second argument.

A fixed shell family supplies unital complex star automorphisms α_j and elements $w_{j,n} \in A$ with $\varphi_j \circ \alpha_j = \omega$, $w_{j,n}^* w_{j,n} = \alpha_j(f_n - f_{n+1})$, and $w_{j,n} w_{j,n}^* = f_n - f_{n+1}$. At the root, α_o is the identity and $w_{o,n} = f_n - f_{n+1}$. Fix $i \in I$ and put $f_{i,n} = \alpha_i(f_n)$. The family is supplied; its existence is not an extra conclusion here. The condition $j \neq i$ means that the unitary-equivalence classes are distinct, not merely that two state functionals in the same class differ. The selected representation π_j is irreducible and has nonzero Hilbert space because $\|\xi_j\| = 1$.

Conclusion

The transported flag leaves no nonzero vector fixed in an inequivalent chosen fiber. As a separate consequence of decreasing orthogonal projections, $\pi_j(f_{i,n})x \rightarrow 0$ in norm for each fixed $x \in H_j$. Thus the surviving line in the matching fiber is sharply separated from every other chosen class.

Irreducibility is used to make a nonzero candidate fixed vector cyclic. The conclusion is not asserted for an arbitrary reducible representation or for a different vector state in the same GNS class. The exact source proves equality of projections; strong convergence uses the additional cited projection theorem and does not imply operator-norm convergence.

Proof route

If F_j contained a unit vector η , the transported CAR compression estimate $\|f_{i,n}bf_{i,n} - \varphi_i(b)f_{i,n}\| \rightarrow 0$, valid for every $b \in A$, would imply $\langle \eta, \pi_j(b)\eta \rangle = \varphi_i(b)$. Irreducibility makes η cyclic. Equality of this vector state with the state of ξ_i yields a unitary intertwiner from π_i to π_j , contradicting their distinct classes.

▼ Proof steps

1. If $P_j \neq 0$, choose z with $P_j z \neq 0$ and normalize $P_j z$ to a unit vector η . Since P_j projects onto the common range, every $\pi_j(f_{i,n})$ fixes η .
2. Take the matrix coefficient of the represented compression error against η . Fixedness and unit norm turn it into the constant $\langle \eta, \pi_j(b)\eta \rangle - \varphi_i(b)$; norm convergence forces this constant to vanish.
3. The orbit closures of ξ_i and η are their full Hilbert spaces. Equality of vector states gives equality of orbit inner products, so the map $\pi_i(b)\xi_i \mapsto \pi_j(b)\eta$ extends to a surjective complex-linear isometry that intertwines the representations. This contradicts $j \neq i$.

Main citations

- Exact common-range theorem and proof
- The fixed shell-family input
- One fixed transported flag
- Norm compression toward the matching state
- Pure-state GNS fibers are irreducible
- Distinct selected classes admit no unitary intertwiner
- Compression excludes a common fixed vector
- Equal cyclic vector states give unitary transport
- Strong convergence to the intersection projection
- The root state bundled with its purity proof

Lean source signature (exact)

```
theorem selected_commonFixedProjection_eq_zero
  (family : RepresentativeShellFamily) {i j :
MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit}
  (hij : j ≠ i) :
  commonFixedProjection (fun n ↦
    MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation completedRootPureState j
      (transportedFlag family i n)) = 0
```

In the source	Mathematical meaning
<code>family; {i j : GNSClass Limit}</code>	The fixed CAR shell family, its flag index i , and the selected GNS representation index j . These indices are unitary-equivalence classes of pure-state GNS representations.
<code>hij : j ≠ i</code>	The representing class differs from the flag class. Distinct state functionals in one class would not satisfy this condition.
<code>transportedFlag family i n</code>	The projection $f_{i,n} = \alpha_i(f_n)$ in A , with the flag still indexed by i .
<code>selectedRepresentation completedRootPureState j (...)</code>	The represented projection $\pi_j(f_{i,n})$ on the different selected fiber H_j .
<code>commonFixedProjection (fun n ↦ ...) = 0</code>	The orthogonal projection P_j onto $F_j = \bigcap_n \text{ran } \pi_j(f_{i,n})$ is the zero operator on H_j , equivalently $F_j = \{0\}$. The quantification is over these selected fibers, not arbitrary reducible representations.

Further source notes: Here `Limit` is A and `completedRootPureState` is the root state ω , bundled with its purity proof. The quantified indices range over chosen GNS classes, not over arbitrary vectors or arbitrary representations.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_zero`

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